

# TWISTED ECKARDT PERIODS, UNIFORM PRILL-EXCEPTIONAL COVERS, AND A PRYM FIXED-PART OBSTRUCTION

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ABSTRACT. I relate two constructions with finite-unitary coefficients: the Eckardt quotient of the Fano surface of a cubic threefold, whose convolution Tannaka group is  $F_4$ , and Raynaud's bundles without theta divisors on curves. For an explicit marked Eckardt cubic, I prove that the twisted period differential is injective on the full seven-dimensional marked Eckardt tangent space. Its second Higgs iterate has rank between 12 and 21 on a nonempty Zariski-open set of characters; the upper bound is forced by the Eckardt reflection. For torsion characters the horizontal Tate line splits off, and the complementary rank-26 summand has period rank 7 at the distinguished complex embedding.

For every  $g \geq 2$ , I then construct a single irreducible finite representation  $\rho_g : \pi_1(\Sigma_g) \rightarrow \mathrm{SU}(g^g)$  which is Prill-exceptional for every marked complex structure and every point. If  $G_g$  is its image, then

$$1 + g^{2g} \leq |G_g| \leq g^{3g}.$$

This gives a fixed topological Galois cover with at least  $1 + g^g$  sections of every fibre divisor and, for  $g \geq 3$ , disproves the generic-global-generation conjecture for unitary local systems. Independently, every Prill-exceptional connected *étale* cover has degree at least  $g + 2$ , and every such Galois cover has degree at least  $1 + (g + 1)^2$ .

Finally, I prove that maximal Siegel rank at one point of a weight-one period map excludes every virtual constant subvariation. Applied to the degree-27 and degree-3 Prym maps, this prevents flat classes on the smooth-Hurwitz parts of special Fano and Eckardt fibres from extending to finite-index constant factors. These double-cover no-finite-orbit cases were already known; no new case of the general Putman–Wieland conjecture is claimed. Neither the weight-two  $F_4$  summand nor failure of generic global generation alone produces a finite monodromy orbit.

## 1. INTRODUCTION

The two starting points of this paper have the same numerical shadow but live in different Hodge-theoretic categories. Let  $F$  be the Fano surface of lines on a smooth cubic threefold with an Eckardt point, let  $E \subset F$  be the elliptic curve parametrizing the lines through that point, and let

$$A = \mathrm{Alb}(F), \quad B = A/E, \quad u : F \longrightarrow B.$$

The quotient contracts  $E$  and satisfies

$$Ru_* \mathbf{C}_F[2] \simeq \delta_X \oplus \delta_0, \quad X = u(F), \tag{1.1}$$

while the convolution Tannaka group of  $\delta_X$  is the simple group  $F_4$ . Under the resulting restriction of the Albert representation,

$$\mathbf{27}|_{F_4} = \mathbf{1} \oplus \mathbf{26}. \tag{1.2}$$

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The first goal is to determine whether the 26-dimensional term in (1.2) varies maximally along the Eckardt locus, rather than merely identifying its fibrewise Tannaka symmetry.

The second starting point is Raynaud's rank- $g^g$  bundle on a genus- $g$  curve. After a determinant-normalized twist, it is a stable degree-zero bundle  $V$  with finite monodromy such that

$$H^0(C, V) = 0, \quad H^0(C, V(p)) \neq 0 \quad \text{for every } p \in C. \quad (1.3)$$

Condition (1.3) is precisely the nontrivial summand required for a fibre divisor on a finite étale cover to move. A priori the finite representation depends on  $C$ . The second goal is to remove that dependence. I call a finite étale cover  $f : X \rightarrow C$  *Prill-exceptional* if  $h^0(X, \mathcal{O}_X(f^{-1}(p))) \geq 2$  for every  $p \in C$ ; a finite representation is Prill-exceptional when its associated summand has the equivalent nonvanishing property in (1.3).

Here are the main results. The precise forms, including coefficient fields and marked-moduli qualifications, are stated in the body of the paper.

**Theorem 1.1** (Eckardt twisted infinitesimal Torelli). *For the marked Eckardt pair*

$$Y_0 = \{x_0^2 x_1 + x_1^3 + x_2^3 + x_3^3 + x_4^3 = 0\} \subset \mathbf{P}^4, \quad p_0 = [1 : 0 : 0 : 0 : 0],$$

*let  $T_E$  be the tangent space to the local moduli space of marked Eckardt pairs at  $(Y_0, p_0)$ . Then  $\dim T_E = 7$ . There is a nonempty Zariski-open set*

$$\Omega \subset \text{Pic}^0(B) \setminus \{0\}$$

*such that, for  $M \in \Omega$  and  $L = u^*M$ , the infinitesimal Higgs map on  $H^2(F, L)$  is injective on  $T_E$ . Its second iterate satisfies*

$$12 \leq \text{rk } \theta_L^{(2)}|_{\text{Sym}^2 T_E} \leq 21.$$

*The set  $\Omega$  contains torsion points of arbitrarily large order. For a torsion point, the horizontal Tate line in (1.1) splits off. The complementary summand has Hodge numbers  $(6, 14, 6)$  at every complex embedding of its cyclotomic field, and period rank 7 along  $T_E$  at the distinguished embedding defined by  $M$ .*

The proof is an exact Jacobian-ring computation. The image

$$I_E = \mu(\text{Sym}^2 T_E)$$

has dimension 27. The reflection identifies the two six-dimensional extreme Hodge spaces and makes the projected multiplication symmetric, which gives the global upper bound  $\binom{6+1}{2} = 21$ . A saturated rank-six limit at the trivial character produces a  $12 \times 12$  determinant  $-2^7 3^6$  and a flattened  $7 \times 7$  determinant  $-2^3 3^7$ . The latter proves injectivity of the first Higgs field by integrability. The rank-six central space is the fibre of a direct-image bundle; it is not the ten-dimensional untwisted space  $H^0(F, K_F)$ .

**Theorem 1.2** (Uniform Raynaud–Prill representation). *Let  $g \geq 2$  and  $r = g^g$ . There is a nontrivial irreducible representation with finite image*

$$\rho_g : \pi_1(\Sigma_g, x_0) \longrightarrow \text{SU}(r)$$

*such that, for every marked compact Riemann surface  $C$  of genus  $g$  and every  $p \in C$ , the associated bundle  $V_{\rho_g, C}$  satisfies*

$$H^0(C, V_{\rho_g, C}) = 0, \quad H^0(C, V_{\rho_g, C}(p)) \neq 0.$$

*If  $G_g = \text{im } \rho_g$ , then*

$$1 + g^{2g} \leq |G_g| \leq g^{3g}.$$

*The fixed topological Galois cover associated with  $\ker \rho_g$  consequently satisfies*

$$h^0(X_C, \mathcal{O}_{X_C}(f_C^{-1}(p))) \geq 1 + g^g$$

for every complex structure and every point.

The uniformization is a finite-type argument, but two details are essential. Schneider’s stability theorem must be applied to the canonical Fourier–Mukai Raynaud bundle together with its theta-group equivariance. On a finite level cover of moduli, the mapping class subgroup must fix each actual homomorphism  $\rho_i$ , not only its conjugacy class; otherwise a projective descent cocycle remains. With these points in place, there are only finitely many possible marked representations, their all-points failure loci are Zariski closed, and irreducibility of the level cover forces one representation to work everywhere.

For  $g \geq 3$ , Theorem 1.2 gives a finite-unitary counterexample to Litt’s generic-global-generation conjecture [18, Conjecture 6.3.5]: the bundle  $V_{\rho_g, C}^\vee \otimes K_C$  is not generated at any point for any complex structure. Since  $|G_g| \geq 1 + g^{2g} > g^2$ , the small-deck-group theorem of Landesman–Litt [16, Theorem 1.2] does not decide the Putman–Wieland orbit question for this cover. Failure of generic global generation alone does not produce a finite orbit.

The same evaluation map gives a numerical constraint that is independent of Raynaud’s construction.

**Theorem 1.3** (Degree bounds). *Let  $C$  have genus  $g \geq 2$ . If a nontrivial stable degree-zero bundle  $W$  of rank  $r$  satisfies  $H^0(C, W(p)) \neq 0$  for every  $p$ , and*

$$c = \min_{p \in C} h^0(C, W(p)),$$

*then  $r \geq gc + 1$ . Consequently, every connected Prill-exceptional finite étale cover of  $C$  has degree at least  $g + 2$ ; in the Galois case the degree is at least  $1 + (g + 1)^2$ .*

The Putman–Wieland connection is global and weight one. The existence of a nonzero finite orbit is equivalent to the existence, after finite étale base change, of a nonzero constant rational subvariation. The following rank obstruction therefore separates special isoperiodic fibres from finite-index monodromy.

**Theorem 1.4** (Maximal-rank fixed-part obstruction). *Let  $S$  be a connected smooth complex quasi-projective variety, and let  $\mathbb{V}_{\mathbf{Z}}$  be a polarized integral variation of Hodge structure of type  $\{(1, 0), (0, 1)\}$  and rank  $2n$ . If its local period map has differential of rank  $n(n + 1)/2$  at one point of  $S$ , then  $\mathbb{V}_{\mathbf{Q}}$  has no nonzero constant subvariation after any connected finite étale base change.*

For the Prym map  $P_6 : \mathcal{R}_6 \rightarrow \mathcal{A}_5$ , both spaces have dimension 15 and the generic degree is 27 [8, Theorem I.2.1]. Theorem 1.4 excludes a finite orbit in its anti-invariant cohomology and gives a no-extension statement for flat classes on the smooth-Hurwitz part of a special Fano fibre. The same argument applies to the degree-3 map  $P_{3,4}$  and its special Eckardt fibre. These are independent period-rank proofs of double-cover cases already covered by [16, Theorem 1.2]; the ramified case belongs to the punctured-surface formulation and is not an unramified cover of a closed genus-three surface.

The limitation is structural. The  $F_4$  calculation gives a nonisotrivial weight-two summand of rank 26 over a cyclotomic field. The uniform Raynaud representation gives failure of generic global generation in weight one. A Putman–Wieland counterexample requires a rational flat class after finite-index base change. Neither construction supplies that additional structure, while Theorem 1.4 rules it out for the maximal-rank Prym maps considered here.

## 2. THE ECKARDT QUOTIENT AND ITS $F_4$ TANNAKA GROUP

All varieties in Sections 2–4 are over  $\mathbf{C}$ . Constructible complexes have complex coefficients. For an integral subvariety  $Z$  of an abelian variety,  $\delta_Z$  denotes its perverse intersection complex,

extended by zero. The convolution Tannaka group  $G(\delta_Z)$  is taken in the quotient by negligible complexes, following [15].

Consider

$$Y_0 = \{F_0 = 0\} \subset \mathbf{P}^4, \quad F_0 = x_0^2 x_1 + x_1^3 + x_2^3 + x_3^3 + x_4^3, \quad (2.1)$$

and  $p_0 = [1 : 0 : 0 : 0 : 0]$ . The five partial derivatives are

$$2x_0 x_1, \quad x_0^2 + 3x_1^2, \quad 3x_2^2, \quad 3x_3^2, \quad 3x_4^2.$$

They have no common projective zero, so  $Y_0$  is smooth. The tangent hyperplane at  $p_0$  is  $x_1 = 0$ , and its section is the cone over the smooth Fermat plane cubic  $x_2^3 + x_3^3 + x_4^3 = 0$ . Thus  $p_0$  is an Eckardt point.

Let  $F = F(Y_0)$  be the Fano surface of lines. Its Albanese variety  $A$  has dimension five, and the Albanese map is an embedding [6]. The lines through  $p_0$  form a smooth elliptic curve  $E \subset F$ . Rouleau's reflection  $\iota : F \rightarrow F$  fixes  $E$  pointwise, acts by  $+1$  on  $T_0 E$ , and by  $-1$  on  $T_0 A/T_0 E$  [24]. Normalize the Albanese embedding by translating by a point of  $E$  so that every coplanar triple of lines has sum zero. This normalization leaves  $E \subset A$  unchanged and gives

$$u \circ \iota = [-1]_B \circ u, \quad B = A/E, \quad (2.2)$$

where  $u : F \rightarrow B$  is the quotient map.

**Proposition 2.1** (The Eckardt  $F_4$  quotient). *Put  $X = u(F) \subset B$ . Then  $u : F \rightarrow X$  is birational, its only positive-dimensional fibre is  $E = u^{-1}(0)$ , and*

$$Ru_* \mathbf{C}_F[2] \simeq \delta_X \oplus \delta_0, \quad \chi(B, \delta_X) = 26. \quad (2.3)$$

Moreover,

$$G(\delta_X) \simeq F_4, \quad \mathbf{27}|_{F_4} \simeq \mathbf{1} \oplus \mathbf{26}. \quad (2.4)$$

Here  $\mathbf{27}$  is the Albert representation of the simply connected group  $E_6$  and  $\mathbf{26}$  is the minimal representation of  $F_4$ .

*Proof.* The tangent-bundle theorem identifies  $\mathbf{P}(T_s F)$  with the line  $\ell_s \subset Y_0$  represented by  $s$ , inside  $\mathbf{P}(T_0 A) = \mathbf{P}^4$  [6]. Since  $\mathbf{P}(T_0 E) = \{p_0\}$  [24, Proposition 4], the differential of  $u$  has a kernel precisely at the points representing lines through  $p_0$ , namely on  $E$ . Thus  $E$  is the only contracted curve.

To compute the generic degree, project the lines  $\ell_s$  from  $p_0$ . If  $u(s) = u(t)$  at a general point outside  $B[2]$ , the tangent-plane identity shows that  $\ell_s$  and  $\ell_t$  lie in the same plane through  $p_0$ . The Eckardt plane-section decomposition says that the only possibilities away from  $E$  are  $t = s$  and  $t = \iota(s)$ . In the second case (2.2) gives  $u(t) = -u(s)$ , forcing  $u(s) \in B[2]$ . Hence the generic degree is one.

The map is semismall. Its unique relevant exceptional stratum is the origin, and  $E^2 = -3$  [24, Proposition 10]. Thus the local intersection form is nondegenerate of rank one. The semismall decomposition theorem [7] therefore gives exactly one skyscraper summand. Since  $\chi(F) = 27$  [6], taking Euler characteristics proves (2.3).

For completeness, I recall the group-theoretic identification. With the coplanar-triple normalization, the incidence correspondence supplies an invariant cubic in the convolution category. Krämer's Gauss-monodromy calculation identifies the connected derived Tannaka group of  $\delta_F$  with  $E_6$  on its minuscule 27-dimensional module [13]; the invariant cubic and its full stabilizer then give  $G(\delta_F) = E_6$  [14]. Functoriality under  $A \rightarrow B$  and (2.3) give a closed subgroup  $H = G(\delta_X) \subset E_6$  with

$$\mathbf{27}|_H = \mathbf{1} \oplus V, \quad \dim V = 26.$$

The translation stabilizer of  $X$  is trivial. To see this, factor  $u$  through the normalization  $F \rightarrow X^\nu \rightarrow X$ . The first map is an isomorphism away from  $E$  and has no other exceptional curve.

The image  $z$  of  $E$  is singular: over a smooth surface point, the exceptional curves of a birational morphism from a smooth surface arise from point blow-ups and are rational, whereas  $E$  is elliptic. Thus  $z$  is the unique singular point of  $X^\nu$ . A translation preserving  $X$  lifts to  $X^\nu$ , fixes  $z$ , and therefore fixes its image  $0 \in B$ ; the translation is zero. Thus  $\delta_X$  is nondivisible in the sense of Javanpeykar–Krämer–Lehn–Maculan. Their irreducibility theorem [10, Proposition 3.3(2)] implies that  $V|_{H^\circ}$  is irreducible.

Let  $v_0$  span the fixed line in the Albert algebra  $J$ . If its cubic norm vanished, then  $v_0$  would have Albert rank one or two. The Hessian of the Albert cubic has kernel dimension respectively 17 or 9 at standard representatives of these two orbits. On the other hand,  $H^\circ$ -equivariance and  $J|_{H^\circ} = \mathbf{1} \oplus V$  allow only the kernel dimensions 0, 1, 26, 27, a contradiction. Thus the norm of  $v_0$  is nonzero. Its stabilizer in  $E_6$  is  $F_4$  [26], so  $H \subset F_4$ . Finally, no proper positive-dimensional closed subgroup of  $F_4$  in characteristic zero acts irreducibly on the minimal 26-dimensional module [17, Theorem 1 and Table 1.2]. Hence  $H^\circ = F_4$ , and  $F_4 = H^\circ \subset H \subset F_4$ . This proves (2.4).  $\square$

*Remark 2.2.* Proposition 2.1 identifies a fibrewise convolution Tannaka group. It does not identify the connected geometric monodromy of a variation with  $F_4$ . Such an identification requires an additional comparison of Hodge tensors or an equivalent normality argument.

### 3. THE MARKED ECKARDT BRANCH AND THE REFLECTION BOUND

Let

$$\iota(x_0, x_1, x_2, x_3, x_4) = (-x_0, x_1, x_2, x_3, x_4).$$

The projective involution of  $Y_0$  induces Roulleau’s reflection on  $F(Y_0)$  by the plane-section characterization, and both involutions are denoted by  $\iota$ . At the highly symmetric cubic (2.1), the unmarked Eckardt locus has several local branches. Throughout this section,  $T_E$  means the tangent space at  $(Y_0, p_0)$  to the local moduli space of *marked* Eckardt pairs, equivalently the local fixed branch determined by  $p_0$ .

The Jacobian ring is

$$R = \mathbf{C}[x_0, \dots, x_4]/(x_0x_1, x_0^2 + 3x_1^2, x_2^2, x_3^2, x_4^2). \quad (3.1)$$

The standard Jacobian-ring identification  $H^1(Y_0, T_{Y_0}) \simeq R_3$ , composed with the deformation isomorphism  $H^1(Y_0, T_{Y_0}) \simeq H^1(F, T_F)$ , gives  $H^1(F, T_F) \simeq R_3$ ; see the proof of [12, Proposition 3.8]. Define

$$\begin{aligned} a_0 &= x_2x_3x_4, & a_1 &= x_1x_3x_4, & a_2 &= x_1x_2x_4, & a_3 &= x_1x_2x_3, \\ a_4 &= x_1^2x_4, & a_5 &= x_1^2x_3, & a_6 &= x_1^2x_2. \end{aligned}$$

**Lemma 3.1.** *Under  $H^1(F, T_F) \simeq R_3$ ,*

$$T_E = R_3^\iota = \langle a_0, \dots, a_6 \rangle.$$

*In particular,  $\dim T_E = 7$ .*

*Proof.* After shrinking, choose an  $\iota$ -equivariant analytic  $\mathrm{PGL}_5$ -slice: the uniqueness of the reflection attached to a marked Eckardt point permits every nearby reflection to be analytically conjugated to the fixed involution  $\iota$ , and the marked point to  $p_0$ . Every marked Eckardt deformation in this slice has an equation of the form

$$x_0^2\ell(x_1, \dots, x_4) + C(x_1, \dots, x_4),$$

where  $\ell \neq 0$  at a smooth marked Eckardt pair. Conversely, every smooth invariant deformation of this form has  $p_0$  as a marked Eckardt point: its tangent section  $\ell = 0$  is the cone with vertex  $p_0$  over  $C|_{\ell=0} = 0$ . Hence the deformation functor of the marked branch is the  $\iota$ -fixed subfunctor. A finite-order fixed locus is smooth in characteristic zero, and its tangent space is  $R_3^\iota$ . Reduction of cubic monomials by (3.1) gives exactly the seven displayed basis elements.  $\square$

Let

$$\mu : \mathrm{Sym}^2 H^1(F, T_F) \longrightarrow H^2(F, \wedge^2 T_F) \quad (3.2)$$

be cup product. Krämer–Litt–Maculan identify the relevant coefficient of  $\mu$  with

$$\nu : \mathrm{Sym}^2 R_3 \longrightarrow \mathrm{Hom}(\wedge^2 R_1, \wedge^2 R_4), \quad \nu(f \odot g)(r \wedge s) = fr \wedge gs + gr \wedge fs. \quad (3.3)$$

This is the Jacobian-ring formula in the proof of [12, Proposition 3.8].

**Proposition 3.2.** *The restricted cup product has kernel*

$$\ker(\mu|_{\mathrm{Sym}^2 T_E}) = \langle a_1 a_6 + a_2 a_5 + a_3 a_4 \rangle. \quad (3.4)$$

Consequently,

$$I_E := \mu(\mathrm{Sym}^2 T_E) \quad \text{has dimension } 27.$$

*Proof.* In the bases of Appendix A, exact row reduction of (3.3) gives the relation in (3.4). The appendix also records a  $27 \times 27$  minor with determinant  $-128$ , so this is the entire kernel of  $\nu|_{\mathrm{Sym}^2 T_E}$ .

It remains to pass from  $\nu$  to  $\mu$ . The binary cubic  $x_0^2 x_1 + x_1^3$  has three distinct roots, so  $Y_0$  is projectively equivalent to the Fermat cubic. At the Fermat cubic, (3.3) has rank 50 [12, Proposition 3.6]. The bicanonical section of lines of second type lies in  $\ker \mu^\vee$ , so  $\mathrm{rk} \mu \leq 50$  [12, Proposition 3.7]. Since  $\nu$  factors through  $\mu$ , both maps have rank 50 and therefore have the same kernel. Restriction to  $\mathrm{Sym}^2 T_E$  proves the proposition.  $\square$

Let  $\widehat{B} = \mathrm{Pic}^0(B)$  and take  $0 \neq M \in \widehat{B}$ . Let  $\mathbb{M}$  be the unique unitary rank-one local system whose holomorphic bundle is  $M$ , and put

$$\mathbb{L} = u^* \mathbb{M}, \quad L = u^* M.$$

For  $0 \leq p \leq 2$ , write

$$H_L^p = H^{2-p}(F, \Omega_F^p \otimes L).$$

Put  $H_L^p = 0$  otherwise. The extreme Hodge spaces satisfy

$$H_L^2 = E_L = H^0(F, K_F \otimes L), \quad (H_L^0)^\vee = E_{L^{-1}} = H^0(F, K_F \otimes L^{-1}).$$

They both have dimension six by [12, Lemma 3.3]. Let

$$\pi_E : H^0(F, K_F^2) \longrightarrow I_E^\vee$$

be the Serre-dual projection.

On a simply connected analytic neighborhood of  $(Y_0, p_0)$  in the marked Eckardt branch, use an Ehresmann trivialization of the pair  $(F, E)$  to transport the character of  $\mathbb{L}$ . It remains trivial on  $E_s$  and therefore is pulled back from  $B_s = \mathrm{Alb}(F_s)/E_s$ . The relative twisted cohomology is then a complex variation of Hodge structure. For  $\xi \in T_E$ , its infinitesimal Higgs field

$$\theta_L(\xi) : H_L^p \longrightarrow H_L^{p-1}$$

is contraction and cup product with the Kodaira–Spencer class  $\xi$ . Its top-to-bottom second iterate factors through (3.2); under Serre duality its coefficient map is the projected multiplication used below, as in [12, Equation (2.1) and the proof of Proposition 2.6].

**Proposition 3.3** (Reflection bound). *For every  $0 \neq M \in \widehat{B}$ , the top-to-bottom second Higgs iterate along  $T_E$  satisfies*

$$\mathrm{rk} \theta_L^{(2)}|_{\mathrm{Sym}^2 T_E} \leq 21. \quad (3.5)$$

*Proof.* Equation (2.2) gives  $\iota^*L \simeq L^{-1}$ . Rigidify  $M$  at the origin. The theorem of the cube gives a rigidified isomorphism  $[-1]^*M \simeq M^{-1}$ ; after pullback it may be normalized so that applying  $\iota$  twice is the identity. Because  $\iota$  fixes  $T_E$  pointwise, equivariance of cup product shows that it fixes  $I_E$  pointwise. Hence  $\pi_E \iota^* = \pi_E$ .

Choose the involutive normalization

$$\rho_L = \iota^* : E_L \xrightarrow{\sim} E_{L^{-1}}, \quad \rho_{L^{-1}} \rho_L = 1,$$

and define

$$b_L(s, t) = \pi_E(s \rho_L(t)).$$

Then

$$b_L(t, s) = \pi_E(\iota^*(t \rho_L(s))) = \pi_E(\rho_L(t)s) = b_L(s, t),$$

so  $b_L$  factors through  $\text{Sym}^2 E_L$ . By [12, Equation (2.1) and the proof of Proposition 2.6], the transpose of the coefficient map of  $\theta_L^{(2)}$  on  $I_E$  is exactly the projected multiplication  $E_L \otimes E_{L^{-1}} \rightarrow I_E^\vee$ . Therefore

$$\text{rk } \theta_L^{(2)}|_{\text{Sym}^2 T_E} = \text{rk } b_L \leq \dim \text{Sym}^2 E_L = 21.$$

□

#### 4. MAXIMAL INFINITESIMAL VARIATION ON THE MARKED BRANCH

The residue action of  $\iota$  includes the determinant character: under  $H^0(F, \Omega_F^1) \simeq R_1$ , the vector  $x_0$  is invariant and  $x_1, \dots, x_4$  are anti-invariant. Let

$$\alpha \in R_1^\vee, \quad \alpha(x_0) = 0, \quad \alpha(x_1) = \alpha(x_2) = \alpha(x_3) = \alpha(x_4) = 1. \quad (4.1)$$

The annihilation of the invariant line means that  $\alpha \in T_0 \hat{B} \subset T_0 \text{Pic}^0(F)$ .

Choose a smooth analytic arc  $\Delta \rightarrow \hat{B}$  through zero with tangent  $\alpha$ , and let  $\mathcal{L}$  be the induced Poincaré line bundle on  $F \times \Delta$ . Write  $p : F \times \Delta \rightarrow \Delta$  and set

$$\mathcal{E} = p_*(K_F \otimes \mathcal{L}), \quad \mathcal{E}^- = p_*(K_F \otimes \mathcal{L}^{-1}). \quad (4.2)$$

**Lemma 4.1** (Saturated central fibre). *The sheaves  $\mathcal{E}$  and  $\mathcal{E}^-$  are locally free of rank six. Their central fibres inject into  $H^0(F, K_F)$  and satisfy*

$$\mathcal{E}_0 = \mathcal{E}_0^- = U_\alpha := \wedge^2 \ker \alpha, \quad \dim U_\alpha = 6. \quad (4.3)$$

*Multiplication of relative sections defines a regular morphism*

$$\mathcal{E} \otimes \mathcal{E}^- \longrightarrow H^0(F, K_F^2) \otimes \mathcal{O}_\Delta. \quad (4.4)$$

*In particular, its central value requires neither a pole nor a renormalization.*

*Proof.* Both direct images are torsion-free on the smooth curve  $\Delta$ , hence locally free. Away from zero they have rank six by [12, Lemma 3.3]. A relative section whose restriction at zero vanishes is divisible by a local parameter, so restriction embeds each central fibre into  $H^0(F, K_F)$ . The first obstruction to extending a canonical form is cup product with  $\alpha$  for  $\mathcal{E}$  and with  $-\alpha$  for  $\mathcal{E}^-$ . Under the residue identifications, [12, Lemma 3.4] identifies this kernel with  $\wedge^2 \ker \alpha$ , which has dimension six. Rank and injectivity give (4.3). Ordinary multiplication on  $F \times \Delta$  gives (4.4); all sheaves involved are locally free, so the morphism is regular at zero. □

Compose (4.4) with  $\pi_E$ , transpose, and precompose with  $\mu|_{\text{Sym}^2 T_E}$ . This gives a morphism of vector bundles

$$\bar{\Theta}^{(2)} : \text{Sym}^2 T_E \otimes \mathcal{O}_\Delta \longrightarrow \text{Hom}(\mathcal{E}, (\mathcal{E}^-)^\vee). \quad (4.5)$$

For  $t \neq 0$ , Serre duality and [12, Equation (2.1)] identify it with the top-to-bottom second Higgs iterate. At zero, it is the saturated limit obtained by restricting ordinary multiplication to  $U_\alpha \otimes U_\alpha$ . Notice that  $\bar{\Theta}_0^{(2)}$  is not the second Higgs field of the untwisted ten-dimensional space  $H^0(F, K_F)$ .

For the exact computation, put

$$z = x_0, \quad w_1 = x_1 - x_4, \quad w_2 = x_2 - x_4, \quad w_3 = x_3 - x_4,$$

and order a basis of  $U_\alpha$  as

$$\begin{aligned} u_0 &= z \wedge w_1, & u_1 &= z \wedge w_2, & u_2 &= z \wedge w_3, \\ u_3 &= w_1 \wedge w_2, & u_4 &= w_1 \wedge w_3, & u_5 &= w_2 \wedge w_3. \end{aligned} \quad (4.6)$$

Formula (3.3) computes the matrix of (4.5) at zero. The complete certificates are recorded in Appendix A.

**Lemma 4.2** (The two certificate minors). *The map  $\overline{\Theta}_0^{(2)}$  has rank exactly 12. Moreover, the flattening*

$$\Phi_0 : T_E \longrightarrow \text{Hom}(T_E, \text{Hom}(U_\alpha, U_\alpha^\vee)), \quad \Phi_0(\xi)(\eta) = \overline{\Theta}_0^{(2)}(\xi\eta), \quad (4.7)$$

*is injective.*

*Proof.* The involution splits  $U_\alpha = U_\alpha^+ \oplus U_\alpha^-$  with dimensions  $3 + 3$ . Since  $I_E$  is fixed, the mixed part of the projected multiplication is zero. At  $t = 0$  the two direct-image fibres coincide with  $U_\alpha$ , and ordinary multiplication is commutative; hence the surviving same-parity blocks factor through  $\text{Sym}^2 U_\alpha^+ \oplus \text{Sym}^2 U_\alpha^-$ . Its rank is therefore at most  $\dim \text{Sym}^2 U_\alpha^+ + \dim \text{Sym}^2 U_\alpha^- = 6 + 6 = 12$ . The matrix in (A.1) is a  $12 \times 12$  minor and has determinant  $-2^7 3^6$ , proving equality.

For (4.7), the seven rows and columns displayed in (A.2) give a determinant  $-2^3 3^7$ . Thus  $\Phi_0$  has rank seven and is injective.  $\square$

**Theorem 4.3** (Maximal variation on the marked Eckardt branch). *There is a nonempty Zariski-open subset*

$$\Omega \subset \widehat{B} \setminus \{0\}$$

*such that, for every  $M \in \Omega$  and  $L = u^*M$ , the full infinitesimal Higgs map*

$$\theta_L|_{T_E} : T_E \longrightarrow \bigoplus_p \text{Hom}(H_L^p, H_L^{p-1}) \quad (4.8)$$

*is injective, and*

$$12 \leq \text{rk } \theta_L^{(2)}|_{\text{Sym}^2 T_E} \leq 21. \quad (4.9)$$

*The open set  $\Omega$  contains torsion points of arbitrarily large order.*

*Proof.* Extend the flattening in (4.7) to  $\Phi_t$  by using (4.5). Lemma 4.2 and openness of matrix rank imply that  $\Phi_t$  is injective and that the second iterate has rank at least 12 for all sufficiently small  $t \neq 0$ . Higgs integrability gives

$$\theta_{L_t}^{(2)}(\xi, \eta) = \theta_{L_t}(\eta)\theta_{L_t}(\xi) = \theta_{L_t}(\xi)\theta_{L_t}(\eta).$$

Therefore  $\ker \theta_{L_t} \subseteq \ker \Phi_t$ , and the first Higgs field is injective along  $T_E$ . Proposition 3.3 gives the upper bound 21.

On  $\widehat{B} \setminus \{0\}$ , [12, Lemma 3.3] and cohomology and base change make the two Poincaré direct images algebraic vector bundles of rank six. Multiplication and flattening are algebraic morphisms. The two rank conditions define a Zariski-open locus; the punctured analytic arc meets it by Lemma 4.2, so it is nonempty. Denote this locus by  $\Omega$ . Torsion points are Zariski dense in an abelian variety. Since the torsion points of bounded order form a finite set, every nonempty open subset contains torsion points of arbitrarily large order.  $\square$

**Corollary 4.4** (The 26-dimensional torsion summand). *Let  $M \in \Omega$  have finite order  $n$ , let  $K = \mathbf{Q}(\zeta_n)$ , and let  $\chi : \pi_1(B) \rightarrow \mu_n \subset K^\times$  be the character determined by  $M$ . Let  $\tau_0 : K \hookrightarrow \mathbf{C}$  be the embedding for which the associated holomorphic line bundle is  $M$ . After passage to a finite level cover and restriction to a simply connected analytic neighborhood  $S_E$  in the local marked*

Eckardt moduli space, write  $\pi : \mathcal{F} \rightarrow S_E$  for the resulting Fano-surface family and  $\mathbb{L}_\chi$  for the transported pullback of  $\chi$ , rigidified to be trivial on the relative elliptic divisor. Then the  $K$ -local system

$$\mathbb{V}_\chi = R^2\pi_*\mathbb{L}_\chi \simeq K(-1) \oplus \mathbb{W}_\chi, \quad \mathrm{rk}_K \mathbb{W}_\chi = 26 \quad (4.10)$$

as a horizontal direct sum. For every embedding  $\tau : K \hookrightarrow \mathbf{C}$ , the Hodge numbers of  $\mathbb{W}_{\chi,\tau} = \mathbb{W}_\chi \otimes_{K,\tau} \mathbf{C}$  are  $(6, 14, 6)$ . For  $\tau = \tau_0$ , its period differential along  $T_E$  has rank seven. Consequently,  $\mathrm{Res}_{K/\mathbf{Q}} \mathbb{W}_\chi$  has  $\mathbf{Q}$ -rank  $26[K : \mathbf{Q}]$  and period differential of rank seven along  $T_E$ .

*Proof.* Let  $i : \mathcal{E} \hookrightarrow \mathcal{F}$  be the relative elliptic divisor. Fibrewise triviality shows that  $\mathbb{L}_\chi|_{\mathcal{E}}$  is pulled back from  $S_E$ ; since  $S_E$  is simply connected, the chosen rigidification makes this restriction trivial. Relative Gysin and restriction are morphisms of variations of Hodge structure

$$K(-1) \xrightarrow{i_*} R^2\pi_*\mathbb{L}_\chi \xrightarrow{i^*} K(-1).$$

Their composite is multiplication by the fibrewise self-intersection  $E_s^2 = -3$ . Hence

$$P = -\frac{1}{3} i_* \circ i^*$$

is a horizontal idempotent. Its image is  $K(-1)$  and its kernel is the rank-26 variation  $\mathbb{W}_\chi$ , proving (4.10).

For every embedding  $\tau$ , the character  $\tau\chi$  is nontrivial. The proof of [12, Lemma 3.3] gives

$$H^j(F, \mathbb{L}_{\chi,\tau}) = 0 \quad (j \neq 2).$$

Thus  $\dim_{\mathbf{C}} H^2(F, \mathbb{L}_{\chi,\tau}) = \chi(F) = 27$ . Its two extreme Hodge numbers are six by the same lemma, so its Hodge numbers are  $(6, 15, 6)$ . Removing the Tate line gives  $(6, 14, 6)$  for every  $\tau$ .

For the distinguished embedding  $\tau_0$ , Theorem 4.3 gives rank seven; the Tate line has zero Higgs field, so this rank occurs on  $\mathbb{W}_{\chi,\tau_0}$ . After restriction of scalars the period differential contains this component and has source  $T_E$  of dimension seven, hence has rank exactly seven.  $\square$

*Remark 4.5.* The rank 26 in Corollary 4.4 is over the character field  $K$ , or over  $\mathbf{C}$  after choosing one embedding. The underlying rational variation has rank  $26[K : \mathbf{Q}]$ , not 26 in general. Moreover, the result is a weight-two statement. These two facts are essential when comparing it with Putman–Wieland.

## 5. RAYNAUD BUNDLES AND FINITE MONODROMY

Let  $C$  be a smooth projective connected curve of genus  $g \geq 2$ , let  $J = \mathrm{Pic}^0(C)$ , and fix a symmetric theta divisor  $\Theta$ . Let  $i : C \hookrightarrow J$  be an Abel–Jacobi embedding. With the conventional harmless inversion depending on the Fourier–Mukai normalization, define the canonical Raynaud bundle

$$\mathcal{E}_n = \mathcal{F}_J(\mathcal{O}_J(-n\Theta))[g]. \quad (5.1)$$

Set  $R_n = i^*\mathcal{E}_n$ . Raynaud proves that  $R_n$  is semistable and

$$[n]^*\mathcal{E}_n \simeq \mathcal{O}_J(n\Theta) \otimes H^0(J, \mathcal{O}_J(n\Theta))^\vee, \quad (5.2)$$

and

$$\mathrm{rk} R_n = n^g, \quad \mu(R_n) = \frac{g}{n}, \quad H^0(C, R_n \otimes \alpha) \neq 0 \quad (\alpha \in \mathrm{Pic}^0(C)). \quad (5.3)$$

The last assertion follows on all of  $\mathrm{Pic}^0(C)$  from Raynaud’s generic assertion by upper semicontinuity [23].

**Lemma 5.1** (Stability). *The bundle  $R_n = i^*\mathcal{E}_n$  is stable.*

*Proof.* This is [25, Corollary 2.2]; I recall why the canonical Fourier–Mukai structure matters. Formula (5.2) is equivariant for Mumford’s theta group of  $\mathcal{O}_J(n\Theta)$ . Its weight-one Schrödinger module  $H^0(J, \mathcal{O}_J(n\Theta))$  is irreducible, and hence so is its dual [25, Proposition 2.1(3)].

The pullback

$$\tilde{C} = C \times_{J, [n]} J$$

is connected because  $i_* : \pi_1(C) \rightarrow \pi_1(J) = H_1(C, \mathbf{Z})$  is abelianization. If a saturated proper subbundle of  $R_n$  had the same slope, its pullback, after tensoring by the inverse of the line factor in (5.2), would be a degree-zero subbundle of a trivial bundle on  $\tilde{C}$ . Such a subbundle is constant: dualize, choose a full set of generating sections, and use degree zero to see that their determinant never vanishes. Descent would therefore give a proper invariant subspace of the Schrödinger module, contradicting irreducibility. Raynaud’s semistability then implies stability.  $\square$

Set

$$r = g^g, \quad R = R_g.$$

Choose  $M \in \text{Pic}^{-1}(C)$  such that

$$M^{\otimes r} \simeq (\det R)^{-1}; \quad (5.4)$$

it exists because multiplication by  $r$  on  $J$  is surjective. Put

$$V_C = R \otimes M. \quad (5.5)$$

Then  $V_C$  is stable of degree zero with trivial determinant. A nonzero section would produce a line subbundle of nonnegative degree, contrary to stability, so

$$H^0(C, V_C) = 0, \quad H^0(C, V_C(p)) \neq 0 \quad (p \in C), \quad (5.6)$$

where the second assertion follows from (5.3) because  $M(p) \in \text{Pic}^0(C)$ .

**Proposition 5.2** (A bounded trivializing cover). *The bundle  $V_C$  is trivialized by a connected finite étale cover of degree at most  $g^{3g}$ . Its Narasimhan–Seshadri representation is irreducible, unitary, has trivial determinant, and has finite image of order at most  $g^{3g}$ .*

*Proof.* Pull  $C \rightarrow J$  back along  $[g] : J \rightarrow J$ . The resulting connected Galois cover

$$q : \tilde{C} \longrightarrow C$$

has degree  $g^{2g}$ : its subgroup is the kernel of  $\pi_1(C) \rightarrow H_1(C, \mathbf{Z}/g)$ . Equations (5.2) and (5.5) give

$$q^* V_C \simeq N^{\oplus r}, \quad N = \tilde{i}^* \mathcal{O}_J(g\Theta) \otimes q^* M.$$

Taking determinants and using  $\det V_C \simeq \mathcal{O}_C$  gives  $N^{\otimes r} \simeq \mathcal{O}_{\tilde{C}}$ . If  $d \mid r$  is the exact order of  $N$ , the algebra

$$\bigoplus_{a=0}^{d-1} N^{-a}$$

defines an étale  $\mu_d$ -torsor trivializing  $N$ . For  $0 < a < d$ , the degree-zero line bundle  $N^a$  is nontrivial and has no section. Therefore

$$H^0\left(\tilde{C}, \bigoplus_{a=0}^{d-1} N^{-a}\right) = \mathbf{C}.$$

Its coordinate algebra has no nontrivial idempotent, so the torsor is connected. The composite cover has degree

$$dg^{2g} \leq g^g g^{2g} = g^{3g}.$$

By the Narasimhan–Seshadri correspondence [21], the stable degree-zero bundle  $V_C$  corresponds to an irreducible unitary representation  $\rho_C$ . Pullback to the displayed cover is the trivial polystable bundle, hence the restriction of  $\rho_C$  to the covering subgroup is trivial. Therefore its image is

finite and its order is at most the degree of the cover. The determinant character of  $\rho_C$  is the rank-one Narasimhan–Seshadri representation corresponding to  $\det V_C \simeq \mathcal{O}_C$ , hence is trivial. Thus  $\rho_C$  takes values in  $\mathrm{SU}(r)$ .  $\square$

## 6. A SINGLE PRILL REPRESENTATION FOR EVERY COMPLEX STRUCTURE

Fix a pointed oriented closed surface  $(\Sigma_g, x_0)$  and write  $\pi = \pi_1(\Sigma_g, x_0)$ . Proposition 5.2 produces, for each marked complex structure, an irreducible finite representation of rank  $r = g^g$  and image order at most  $g^{3g}$ . I now show that one representation works for all complex structures.

**Lemma 6.1** (Finiteness of the possible representations). *Up to  $\mathrm{GL}_r(\mathbf{C})$ -conjugacy, there are only finitely many nontrivial irreducible representations*

$$\rho : \pi \longrightarrow \mathrm{SL}_r(\mathbf{C})$$

whose image has order at most  $g^{3g}$ .

*Proof.* There are finitely many groups of bounded order: on a set of cardinality  $m$ , there are only finitely many multiplication tables. A homomorphism from the  $2g$ -generated group  $\pi$  to a fixed finite group  $G$  is determined by the images of the generators, so there are at most  $|G|^{2g}$  such homomorphisms. Finally, a finite group has finitely many irreducible complex representations. Retaining only faithful representations of dimension  $r$  with trivial determinant proves the assertion.  $\square$

Average a positive Hermitian form over each finite image and choose representatives

$$\rho_1, \dots, \rho_N : \pi \longrightarrow \mathrm{SU}(r) \tag{6.1}$$

of these conjugacy classes. Let  $\Gamma_{g,1}$  be the pointed mapping class group, acting on  $\pi$  by actual automorphisms. Since each  $\rho_i$  takes values in a fixed finite matrix group, its orbit under precomposition is finite. Define the exact stabilizer

$$\Gamma_i = \{\gamma \in \Gamma_{g,1} : \rho_i \circ \gamma_* = \rho_i\}.$$

Choose  $\ell \geq 3$  and put

$$\Gamma' = \Gamma(\ell) \cap \bigcap_{i=1}^N \Gamma_i. \tag{6.2}$$

Then  $\Gamma'$  is torsion-free and of finite index. The quotient of pointed Teichmüller space

$$S = T_{g,1}/\Gamma'$$

is a connected smooth quasi-projective variety, finite étale over a fine level cover of  $\mathcal{M}_{g,1}$ . It carries a universal pointed curve

$$\pi_{\mathcal{C}} : \mathcal{C} \longrightarrow S. \tag{6.3}$$

The section splits the homotopy sequence and gives

$$\pi_1(\mathcal{C}) \simeq \pi \rtimes \Gamma'.$$

Because (6.2) fixes  $\rho_i$  as an actual homomorphism, not merely up to conjugacy,

$$\tilde{\rho}_i(\delta, \gamma) = \rho_i(\delta) \quad (\delta \in \pi, \gamma \in \Gamma') \tag{6.4}$$

is a homomorphism. It defines a finite local system. By the Riemann existence theorem the corresponding finite topological cover is algebraic, and finite-étale descent of its trivial bundle gives an algebraic flat bundle  $\mathcal{V}_i$  on  $\mathcal{C}$ . This exact descent is the reason for using the pointed mapping class group and the exact stabilizers in (6.2).

Put

$$\mathcal{F}_i = \mathcal{V}_i^{\vee} \otimes \omega_{\mathcal{C}/S}.$$

Every fibre  $V_{i,s}$  is a nontrivial irreducible unitary bundle of degree zero, so  $H^0(V_{i,s}) = 0$ . Serre duality and Riemann–Roch give

$$H^1(F_{i,s}) = 0, \quad h^0(F_{i,s}) = r(g-1).$$

Cohomology and base change make

$$\mathcal{H}_i = (\pi_C)_* \mathcal{F}_i$$

a vector bundle, and there is an evaluation map

$$\varepsilon_i : \pi_C^* \mathcal{H}_i \longrightarrow \mathcal{F}_i. \quad (6.5)$$

Let  $D_i \subset \mathcal{C}$  be the determinantal closed subscheme defined by the maximal minors of  $\varepsilon_i$ ; its support is the nonsurjectivity locus.

**Lemma 6.2** (The all-points locus is closed). *The subset*

$$S_i = \{s \in S : C_s \subseteq D_i\}$$

*is Zariski closed. A point  $s$  lies in  $S_i$  precisely when*

$$H^0(C_s, V_{i,s}(p)) \neq 0 \quad \text{for every } p \in C_s.$$

*Proof.* For  $p \in C_s$ , the cohomology sequence of  $0 \rightarrow F_{i,s}(-p) \rightarrow F_{i,s} \rightarrow (F_{i,s})_p \rightarrow 0$ , followed by Serre duality, gives

$$\text{coker}[H^0(C_s, F_{i,s}) \rightarrow (F_{i,s})_p]^\vee \simeq H^0(C_s, V_{i,s}(p)). \quad (6.6)$$

Thus the asserted pointwise equivalence holds. The morphism  $D_i \rightarrow S$  is proper. Upper semicontinuity of fibre dimension shows that  $\{s : \dim(D_i)_s \geq 1\}$  is closed. Since  $C_s$  is an irreducible curve, a closed subset of  $C_s$  has dimension one exactly when its support is all of  $C_s$ . This closed set is  $S_i$ .  $\square$

**Theorem 6.3** (Uniform Raynaud–Prill theorem). *Let  $g \geq 2$  and  $r = g^g$ . There is a nontrivial irreducible finite representation*

$$\rho_g : \pi_1(\Sigma_g, x_0) \longrightarrow \text{SU}(r) \quad (6.7)$$

*such that, for every marked compact Riemann surface  $C$  of genus  $g$  and every  $p \in C$ ,*

$$H^0(C, V_{\rho_g, C}) = 0, \quad H^0(C, V_{\rho_g, C}(p)) \neq 0. \quad (6.8)$$

*If  $G_g = \text{im } \rho_g$ , then*

$$1 + g^{2g} \leq |G_g| \leq g^{3g}. \quad (6.9)$$

*Proof.* For each  $s \in S$ , the construction in Section 5 and Proposition 5.2 supply an index  $i$  for which  $V_{i,s}$  is isomorphic to  $V_{C_s}$  in (5.5). Hence (5.6) and Lemma 6.2 give

$$S = \bigcup_{i=1}^N S_i.$$

The smooth connected variety  $S$  is irreducible. A finite union of proper closed subsets cannot equal an irreducible variety, so  $S_j = S$  for some  $j$ . Pulling back to Teichmüller space shows that the single marked representation  $\rho_g = \rho_j$  has (6.8) for every marked complex structure.

Every representation in the finite list has image order at most  $g^{3g}$ , proving the upper bound in (6.9); the determinant-one condition was built into the list. By definition  $G_g$  acts faithfully and irreducibly in dimension  $r$ . Since  $r > 1$ , the sum-of-squares formula for irreducible representations of a finite group contains both this representation and the distinct trivial representation, so

$$|G_g| = \sum_{\tau \in \text{Irr}(G_g)} (\dim \tau)^2 \geq 1 + r^2 = 1 + g^{2g}.$$

$\square$

**Corollary 6.4** (A fixed topological Prill cover). *Let  $f_C : X_C \rightarrow C$  be the connected Galois cover associated with  $\ker \rho_g$ . It has the same topological covering type for every marked complex structure, degree  $|G_g|$ , and*

$$h^0(X_C, \mathcal{O}_{X_C}(f_C^{-1}(p))) \geq 1 + g^g \quad (p \in C). \quad (6.10)$$

*Proof.* The regular  $G_g$ -module contains the trivial representation once and the irreducible representation (6.7) with multiplicity  $r$ . Therefore

$$(f_C)_* \mathcal{O}_{X_C} \supset \mathcal{O}_C \oplus V_{\rho_g, C}^{\oplus r}.$$

Projection formula and (6.8) give

$$h^0(X_C, \mathcal{O}_{X_C}(f_C^{-1}(p))) \geq h^0(C, \mathcal{O}_C(p)) + r h^0(C, V_{\rho_g, C}(p)) \geq 1 + r.$$

□

**Corollary 6.5** (Failure of generic global generation). *For every  $g \geq 3$ , the finite-unitary representation  $\rho_g^\vee$  contradicts [18, Conjecture 6.3.5]. More precisely, for every marked complex structure  $C$ , the bundle*

$$V_{\rho_g, C}^\vee \otimes K_C$$

*is not generated by global sections at any point of  $C$ .*

*Proof.* Equation (6.6), with  $V = V_{\rho_g, C}$ , identifies the dual of the evaluation cokernel at  $p$  with  $H^0(C, V(p))$ . This space is nonzero at every point by Theorem 6.3. The ggg conjecture asserts generic global generation for the holomorphic bundle associated with any fixed unitary representation on a generic complex structure. The dual representation fails that assertion for every complex structure. □

## 7. NUMERICAL CONSTRAINTS ON PRILL-EXCEPTIONAL COVERS

The next estimate uses only stability, evaluation, and the vector-bundle Clifford inequality. It applies well beyond the Raynaud construction.

**Theorem 7.1** (Rank–corank inequality). *Let  $W$  be a nontrivial stable bundle of degree zero and rank  $r$  on a curve  $C$  of genus  $g \geq 2$ . Suppose*

$$H^0(C, W(p)) \neq 0 \quad (p \in C),$$

*and put*

$$c = \min_{p \in C} h^0(C, W(p)).$$

*Then*

$$r \geq gc + 1. \quad (7.1)$$

*Proof.* Set  $F = K_C \otimes W^\vee$ . Then  $F$  is stable of rank  $r$ , degree  $2r(g-1)$ , and

$$h^0(F) = r(g-1),$$

because  $H^1(F)^\vee = H^0(W) = 0$ . Let  $U \subset F$  be the saturation of the image of the evaluation map

$$H^0(F) \otimes \mathcal{O}_C \longrightarrow F,$$

and write  $D = \deg U$ . By upper semicontinuity,  $h^0(C, W(p))$  takes its minimum value  $c$  on a nonempty open subset of  $C$ . At a point of this subset, (6.6) says that the evaluation map to the  $r$ -dimensional fibre  $F_p$  has cokernel of dimension  $c$ . Its generic rank is therefore  $r - c$ , and saturation does not change generic rank; hence

$$\mathrm{rk} U = r - c.$$

All sections of  $F$  factor through  $U$ , and  $U \subset F$ , so

$$h^0(U) = h^0(F) = r(g-1). \quad (7.2)$$

The bundle  $U$  is generically globally generated. Every quotient bundle of  $U$  is therefore generically globally generated, and the determinant of such a quotient has a nonzero section; hence every Harder–Narasimhan slope of  $U$  is nonnegative. The maximal-slope Harder–Narasimhan subbundle of  $U$  is a proper subbundle of the stable bundle  $F$ , so its slope, and hence every Harder–Narasimhan slope of  $U$ , is strictly less than  $2g-2$ . Apply the vector-bundle Clifford inequality

$$h^0(Q) \leq \frac{\deg Q}{2} + \operatorname{rk} Q$$

to each semistable Harder–Narasimhan factor and add; see [1, Theorem 2.1]. Equation (7.2) yields

$$D \geq 2r(g-2) + 2c. \quad (7.3)$$

On the other hand, stability of  $F$  and integrality of degree give the strict bound

$$D \leq 2(r-c)(g-1) - 1. \quad (7.4)$$

Combining (7.3) and (7.4) gives  $2cg \leq 2r-1$ . Thus  $r > gc$ , which is (7.1).  $\square$

**Corollary 7.2** (Degree bounds for covers). *Let  $f : X \rightarrow C$  be a connected finite étale cover of a genus- $g$  curve,  $g \geq 2$ , and assume*

$$h^0(X, \mathcal{O}_X(f^{-1}(p))) \geq 2 \quad (p \in C).$$

*Then*

$$\deg f \geq g+2. \quad (7.5)$$

*If  $f$  is Galois, then*

$$\deg f \geq 1 + (g+1)^2. \quad (7.6)$$

*Proof.* By projection formula,

$$h^0(X, \mathcal{O}_X(f^{-1}(p))) = h^0(C, f_*\mathcal{O}_X \otimes \mathcal{O}_C(p)).$$

The finite permutation local system decomposes as

$$f_*\mathcal{O}_X \simeq \mathcal{O}_C \oplus \bigoplus_j W_j^{\oplus m_j},$$

where the  $W_j$  are nontrivial stable finite-unitary bundles. Since  $g \geq 2$ , one has  $h^0(\mathcal{O}_C(p)) = 1$ . Upper semicontinuity makes each locus

$$Z_j = \{p \in C : H^0(C, W_j(p)) \neq 0\}$$

closed. The hypothesis and the displayed decomposition show that these finitely many loci cover  $C$ . Irreducibility of  $C$  gives  $Z_j = C$  for some  $j$ . Theorem 7.1, with  $c \geq 1$ , gives  $\operatorname{rk} W_j \geq g+1$ , and

$$\deg f = \operatorname{rk}(f_*\mathcal{O}_X) \geq 1 + \operatorname{rk} W_j \geq g+2.$$

If  $f$  is Galois, set  $r_j = \operatorname{rk} W_j$ . Its regular representation contains  $W_j$  with multiplicity  $r_j$ . Hence

$$\deg f \geq 1 + r_j^2 \geq 1 + (g+1)^2,$$

because  $r_j \geq g+1$ .  $\square$

The same evaluation quotient measures a common infinitesimal Higgs kernel. This gives a precise bridge to the local linear algebra used in Putman–Wieland arguments.

**Proposition 7.3** (Common Higgs kernel). *In the notation of Theorem 7.1, let  $Q = F/U$  and define*

$$K_W^{1,0} = \bigcap_{\xi \in H^1(C, T_C)} \ker[\theta_\xi : H^0(C, K_C \otimes W) \longrightarrow H^1(C, W)].$$

Then

$$K_W^{1,0} \simeq H^0(C, Q^\vee \otimes K_C^2) \quad (7.7)$$

and

$$\dim K_W^{1,0} \geq \max\{0, c(3g-1) - 2r\}. \quad (7.8)$$

*Proof.* Under Serre duality, the adjoint of the infinitesimal period map is trace multiplication

$$H^0(K_C \otimes W) \otimes H^0(K_C \otimes W^\vee) \longrightarrow H^0(K_C^2).$$

More explicitly, for  $s \in H^0(K_C \otimes W)$ ,  $t \in H^0(K_C \otimes W^\vee)$ , and  $\xi \in H^1(T_C)$ ,

$$\langle \theta_\xi(s), t \rangle_{\text{SD}} = \langle \xi, \text{tr}(s \otimes t) \rangle_{\text{SD}}.$$

Thus a section in the first factor is killed for every tangent vector precisely when the corresponding homomorphism  $F = K_C \otimes W^\vee \rightarrow K_C^2$  vanishes on the evaluation image. It then vanishes on its saturation  $U$ , because a torsion sheaf has no nonzero map to the line bundle  $K_C^2$ . It therefore factors uniquely through  $Q$ , proving (7.7); the converse follows from the same Serre-duality identity.

Since  $\text{rk } Q = c$  and  $\deg Q = 2r(g-1) - D$ , Riemann–Roch gives

$$\chi(Q^\vee \otimes K_C^2) = D + (3c - 2r)(g-1).$$

Using (7.3), this is at least  $c(3g-1) - 2r$ . Since  $h^0 \geq \chi$ , equation (7.8) follows.  $\square$

*Remark 7.4.* Proposition 7.3 is a fibrewise first-order statement. It neither makes  $K_W^{1,0}$  Gauss–Manin horizontal nor gives it parabolic degree zero or a rational structure. A Putman–Wieland finite orbit would instead produce, after finite level, a nonzero rational flat sublocal system; in Higgs terms, a degree-zero horizontal summand. The proposition supplies none of these global properties.

## 8. FINITE ORBITS AND MAXIMAL-RANK PRYM MAPS

I first record the exact dictionary between finite monodromy orbits and constant Hodge-theoretic factors. Let  $S$  be a connected smooth complex quasi-projective variety, let  $\mathbb{V}_{\mathbf{Z}}$  be a polarizable integral variation of Hodge structure on  $S$ , and write  $\Gamma = \pi_1(S, s)$ .

**Proposition 8.1** (Finite orbit dictionary). *The following conditions are equivalent:*

- (i)  $\Gamma$  has a nonzero finite orbit in  $V_{s, \mathbf{C}}$ ;
- (ii) after a connected finite étale base change  $T \rightarrow S$ , the pulled-back local system has a nonzero global flat section;
- (iii) after a connected finite étale base change,  $\mathbb{V}_{\mathbf{Q}}$  contains a nonzero constant  $\mathbf{Q}$ -subvariation of Hodge structure.

*Proof.* A vector has finite orbit exactly when its stabilizer has finite index. By the Riemann existence theorem, the corresponding finite topological cover of  $S$  is algebraic and finite étale; taking the connected component determined by the stabilizer gives (ii), and the converse is immediate.

If a finite-index subgroup  $\Delta$  fixes a nonzero complex vector, then

$$V_{\mathbf{C}}^\Delta = V_{\mathbf{Q}}^\Delta \otimes_{\mathbf{Q}} \mathbf{C} \neq 0,$$

because invariants of a rational representation commute with scalar extension. The theorem of the fixed part makes the maximal trivial sublocal system of a polarizable variation a constant subvariation of Hodge structure [9, Section 6(a)]. This proves (ii) $\Rightarrow$ (iii); the reverse implication is immediate.  $\square$

The same dictionary holds for a smooth quasi-projective Deligne–Mumford stack admitting a finite étale surjective cover by a smooth quasi-projective variety: pull any alleged finite orbit or constant subvariation to a connected component of such a cover and apply Proposition 8.1 there.

**Theorem 8.2** (Maximal-rank fixed-part obstruction). *Let  $S$  be a connected smooth complex quasi-projective variety and let  $\mathbb{V}_{\mathbf{Z}}$  be a polarizable integral variation of Hodge structure of type  $\{(1, 0), (0, 1)\}$  and rank  $2n$ . Suppose that at some  $s \in S$  the differential of its local period map has rank*

$$\mathrm{rk}(d\Phi_s) = \frac{n(n+1)}{2}. \quad (8.1)$$

*Then, for every connected finite étale map  $q : T \rightarrow S$ , the variation  $q^*\mathbb{V}_{\mathbf{Q}}$  contains no nonzero constant subvariation. Equivalently, neither the monodromy of  $\mathbb{V}_{\mathbf{C}}$  nor that of any such pullback has a nonzero finite orbit.*

*The same statement holds for a smooth Deligne–Mumford stack admitting a finite étale surjective cover by a smooth quasi-projective variety.*

*Proof.* Suppose that a constant subvariation  $0 \neq \mathbb{U}_{\mathbf{Q}} \subset q^*\mathbb{V}_{\mathbf{Q}}$  exists. Polarizability gives an orthogonal decomposition

$$q^*\mathbb{V}_{\mathbf{Q}} = \mathbb{U}_{\mathbf{Q}} \oplus \mathbb{W}_{\mathbf{Q}}.$$

Write  $\mathrm{rk}_{\mathbf{Q}} \mathbb{U} = 2a$ , where  $1 \leq a \leq n$ . On Hodge bundles,

$$E^{1,0} = U^{1,0} \oplus W^{1,0}, \quad \mathrm{rk} U^{1,0} = a, \quad \mathrm{rk} W^{1,0} = n - a.$$

The constant summand has zero Higgs field. The two summands are flat and orthogonal, so there are no off-diagonal Higgs terms. Consequently the differential of the period map takes values in

$$\mathrm{Sym}^2((W^{1,0})^\vee) \subset \mathrm{Sym}^2((E^{1,0})^\vee),$$

and at every point of  $T$ ,

$$\mathrm{rk} d(\Phi \circ q) \leq \frac{(n-a)(n-a+1)}{2} < \frac{n(n+1)}{2}. \quad (8.2)$$

Choose  $t \in T$  above the point  $s$  in (8.1). Since  $q$  is étale,  $d(\Phi \circ q)_t$  has rank  $n(n+1)/2$ , contradicting (8.2). Proposition 8.1, applied also after further finite covers, gives the finite-orbit formulation. For a Deligne–Mumford stack, pass to a smooth finite level cover and apply the variety case.  $\square$

In particular, if  $\dim S = n(n+1)/2$  and the classifying morphism  $S \rightarrow \mathcal{A}_{n,\delta}$  is dominant and generically finite, then the hypothesis holds on a nonempty open subset. Maximal rank, rather than generic finiteness onto an arbitrary image, is essential: the map  $B \mapsto A_0 \times B$  may be generically finite onto a product locus while the associated variation contains the constant factor  $H^1(A_0, \mathbf{Q})$ .

**8.1. The degree-27 Prym map.** Let  $\mathcal{R}_6$  be the stack of connected étale double covers

$$\tilde{C} \longrightarrow C, \quad g(C) = 6,$$

and let

$$P_6 : \mathcal{R}_6 \longrightarrow \mathcal{A}_5$$

be the Prym map. The anti-invariant local system

$$\mathbb{V}_6^- = (R^1 \tilde{f}_* \mathbf{Q})^-$$

has rank ten and is the weight-one variation classified by  $P_6$ . One has

$$\dim \mathcal{R}_6 = 15 = \dim \mathcal{A}_5,$$

and Donagi–Smith proved that  $P_6$  is dominant and generically finite of degree 27 [8, Theorem I.2.1]. Adding full level  $N \geq 3$  to the curve and Prym lattice gives a smooth quasi-projective finite étale cover, so Theorem 8.2 applies to this stack.

**Corollary 8.3** (Virtual Prym fixed parts in genus six). *For every connected finite étale base change  $\mathcal{T} \rightarrow \mathcal{R}_6$ , the variation  $\mathbb{V}_6^-|_{\mathcal{T}}$  has no nonzero constant rational subvariation. Equivalently, the anti-invariant Prym monodromy has no nonzero finite orbit.*

*For a fixed connected étale double cover of a genus-six surface, the full liftable mapping-class-group action on  $H^1$  has no nonzero finite orbit. Thus the cover satisfies the Putman–Wieland no-finite-orbit conclusion in the liftable-stabilizer formulation.*

*Proof.* The first assertion is Theorem 8.2. For the second, the rational deck decomposition is

$$H^1(\tilde{C}, \mathbb{Q}) = H^1(\tilde{C}, \mathbb{Q})^+ \oplus H^1(\tilde{C}, \mathbb{Q})^-.$$

The minus part is covered by the first assertion. The plus part is pulled back from  $H^1(C, \mathbb{Q})$ . The stabilizer of the defining nonzero two-torsion class is finite index in the mapping class group, and its image in  $\mathrm{Sp}_{12}(\mathbb{Z})$  is finite index. The standard symplectic representation has no nonzero vector fixed by a finite-index subgroup. Thus neither summand has a nonzero finite orbit.  $\square$

This is not a new case of the Putman–Wieland conjecture: it already follows from [16, Theorem 1.2], because the deck group has order  $2 < 6^2$ . The result above is an independent maximal-period-rank proof and will be used for the no-extension interpretation of special Prym fibres.

At  $(C, \eta)$ , where  $0 \neq \eta \in \mathrm{Pic}^0(C)[2]$ , the codifferential is

$$(dP_6)^* : \mathrm{Sym}^2 H^0(C, K_C \otimes \eta) \longrightarrow H^0(C, K_C^2). \quad (8.3)$$

Both sides have dimension 15, and the degree theorem implies that (8.3) is an isomorphism on a dense open subset; see also [3, Proposition 7.5].

**Corollary 8.4** (Special Prym flat sections do not extend). *Let  $Z \rightarrow \mathcal{R}_6$  be a connected smooth family contained in a fibre of  $P_6$ , for example a smooth–Hurwitz open of the Fano-surface fibre over the intermediate Jacobian of a cubic threefold [5, Theorem 4.1]. Let  $\mathcal{T} \rightarrow \mathcal{R}_6$  be connected and finite étale, and let  $Z'$  be a connected component of  $Z \times_{\mathcal{R}_6} \mathcal{T}$ . No nonzero flat section of  $\mathbb{V}_6^-|_{Z'}$  extends to a global flat section of  $\mathbb{V}_6^-|_{\mathcal{T}}$ .*

*Proof.* An extension would be a nonzero monodromy-invariant vector on  $\mathcal{T}$ , contrary to Corollary 8.3.  $\square$

The proper allowable-cover fibre is the Fano surface, whereas its intersection with the smooth Hurwitz stack is only a Zariski-open part. Corollary 8.4 makes no assertion about subvariations supported entirely on the boundary.

**8.2. The ramified degree-3 analogue.** Let  $\mathcal{R}_{3,4}$  parametrize triples  $(C, \eta, D)$  with  $g(C) = 3$ ,  $D$  reduced of degree four, and  $\eta^{\otimes 2} \simeq \mathcal{O}_C(D)$ . The corresponding double cover is ramified at four points. Its Prym has dimension four and polarization type  $\delta = (1, 2, 2, 2)$ , up to polarized duality. The Prym map

$$P_{3,4} : \mathcal{R}_{3,4} \longrightarrow \mathcal{A}_{4,\delta}$$

has equal source and target dimensions 10 and is dominant of generic degree three [20, 2]; see also [5, Corollary 0.3]. Adding full level  $N \geq 3$  gives a smooth quasi-projective finite étale cover, so Theorem 8.2 applies.

**Corollary 8.5** (Ramified Prym fixed parts). *After no connected finite étale base change of  $\mathcal{R}_{3,4}$  does its anti-invariant Prym variation contain a nonzero constant rational subvariation; equivalently, its monodromy has no nonzero finite orbit.*

*For a fixed branched double cover  $\tilde{C} \rightarrow C$  in this Hurwitz component, the full liftable-stabilizer action on  $H^1(\tilde{C}, \mathbf{Q})$  has no nonzero finite orbit.*

*Let  $E^\circ$  be the smooth-Hurwitz open of the special elliptic fibre in [5, Theorem 5.1]. If  $\mathcal{T} \rightarrow \mathcal{R}_{3,4}$  is connected and finite étale and  $E'$  is a connected component of  $E^\circ \times_{\mathcal{R}_{3,4}} \mathcal{T}$ , then no nonzero flat anti-invariant Prym section on  $E'$  extends to a global flat section on  $\mathcal{T}$ .*

*Proof.* The first assertion follows from maximal generic rank and Theorem 8.2. For the full cohomology, use the rational deck decomposition

$$H^1(\tilde{C}, \mathbf{Q}) = H^1(\tilde{C}, \mathbf{Q})^+ \oplus H^1(\tilde{C}, \mathbf{Q})^-.$$

The minus part is the Prym variation. The plus part is  $H^1(C, \mathbf{Q})$ . The forgetful map from the pure four-punctured mapping class group onto the genus-three mapping class group is surjective; therefore the image of the finite-index liftable stabilizer is finite index, as is its image in  $\mathrm{Sp}_6(\mathbf{Z})$ . The standard representation has no nonzero finite orbit. Finally, an extension of a section from  $E'$  would contradict the first assertion.  $\square$

This double-cover no-finite-orbit case was already known from [16, Theorem 1.2], since the deck group has order  $2 < 3^2$ . The corollary records the independent Prym-period proof and its consequence for the special smooth-Hurwitz fibre; it makes no assertion about boundary-supported subvarieties in the allowable-cover compactification.

The codifferential at  $(C, \eta, D)$  is

$$(dP_{3,4})^* : \mathrm{Sym}^2 H^0(C, K_C \otimes \eta) \longrightarrow H^0(C, K_C^2(D)). \quad (8.4)$$

Both spaces have dimension ten, so (8.4) is an isomorphism at a general point; see [5, Section 5.4]. Removing the four branch points gives an unramified double cover of a four-punctured genus-three surface, and filling the punctures upstairs recovers the original genus-seven covering curve. Thus the anti-invariant variation is a punctured-surface higher-Prym variation in the sense of Putman–Wieland. It is not the variation of an unramified cover of the closed genus-three surface. The statement concerns the finite-index liftable stabilizer, not necessarily a characteristic subgroup acted on by the entire mapping class group.

**8.3. The exact boundary of the comparison.** For the geometric higher-Prym local system on a connected Hurwitz component, a Putman–Wieland finite orbit, a flat section after finite level, and a constant rational subvariation are equivalent by Proposition 8.1. For the curve-cover variation considered by Litt, [18, Proposition 6.3.2] says that generic global generation precludes such a virtual section. Its contraposition gives the one-way implication

$$\begin{aligned} \text{finite orbit} &\implies \text{nonzero rational flat section after finite level} \\ &\implies \text{failure of generic global generation.} \end{aligned} \quad (8.5)$$

Theorem 6.3 establishes the last condition, but the implication cannot be reversed: it produces neither condition to its left. Its deck group satisfies  $|G_g| \geq 1 + g^{2g} > g^2$ , so [16, Theorem 1.2] does not settle this particular higher-Prym representation. Conversely, Corollaries 8.3 and 8.5 exclude the middle condition for the full Prym variations despite the presence of special positive-dimensional fibres.

The 26-dimensional summand in Corollary 4.4 is a weight-two variation on Fano surfaces, defined over a cyclotomic character field. A Putman–Wieland finite orbit in complex weight-one cohomology forces, after finite level, the existence of a nonzero rational flat class. The rank-seven period calculation proves that the distinguished complex component of the  $F_4$  summand is not

isotrivial along the marked Eckardt branch; it does not change its weight or supply the missing rational flat class. This is the precise relationship established here between the  $F_4$ , Raynaud–Prill, and Putman–Wieland constructions.

#### APPENDIX A. EXACT JACOBIAN-RING CERTIFICATES

All entries below are obtained in the Artinian ring (3.1); hence the certificates are computations over  $\mathbf{Q}$ . They are included in full so that the rank assertions do not depend on numerical linear algebra. Products  $a_i a_j$  and  $u_k u_l$  denote the unnormalized products in the corresponding symmetric algebras; no factor  $1/2$  is inserted.

**A.1. The rank-27 restricted cup product.** Use the basis

$$R_4 = \langle x_0 x_2 x_3 x_4, x_1 x_2 x_3 x_4, x_1^2 x_3 x_4, x_1^2 x_2 x_4, x_1^2 x_2 x_3 \rangle$$

and the socle generator  $x_1^2 x_2 x_3 x_4$ . The latter makes the perfect pairing  $R_4 \otimes R_1 \rightarrow R_5 \simeq \mathbf{Q}$  explicit and hence identifies  $\wedge^2 R_4$  with  $(\wedge^2 R_1)^\vee$ . This identification is used for the central and flattened matrices; the  $27 \times 27$  certificate instead uses direct  $\wedge^2 R_4$ -coordinates.

Order the standard basis of  $\wedge^2 R_1$ , and the basis of  $\wedge^2 R_4$  induced by the displayed  $R_4$ -basis, by

$$01, 02, 03, 04, 12, 13, 14, 23, 24, 34.$$

Thus a row  $(ij, kl)$  records the  $kl$ -coordinate of  $\nu(\text{cdot})(x_i \wedge x_j)$  in  $\wedge^2 R_4$ . Take all columns  $a_i a_j$ ,  $0 \leq i \leq j \leq 6$ , except  $a_3 a_4$ , and the following rows of the matrix of  $\nu$ :

$$\begin{aligned} &(01, 01), (01, 02), (01, 03), (01, 04), (02, 03), (02, 04), (03, 04), \\ &(12, 12), (12, 13), (12, 14), (12, 23), (12, 24), (12, 34), \\ &(13, 13), (13, 14), (13, 23), (13, 24), (13, 34), \\ &(14, 14), (14, 24), (14, 34), \\ &(23, 23), (23, 24), (23, 34), (24, 24), (24, 34), (34, 34). \end{aligned}$$

The resulting  $27 \times 27$  determinant is  $-128$ . Direct substitution in (3.3) gives

$$\nu(a_1 a_6 + a_2 a_5 + a_3 a_4) = 0.$$

These two facts prove (3.4).

**A.2. The central rank-12 minor.** With columns

$$(u_0^2, u_0 u_1, u_0 u_2, u_1^2, u_1 u_2, u_2^2; \\ u_3^2, u_3 u_4, u_3 u_5, u_4^2, u_4 u_5, u_5^2)$$

and rows

$$(a_0^2, a_0 a_1, a_0 a_2, a_0 a_4, a_0 a_5, a_0 a_6, a_1^2, a_1 a_2, a_1 a_4, a_1 a_6, a_2^2, a_4^2),$$

where rows are  $T_E$ -products and columns are  $U_\alpha$ -products, the relevant minor is

$$\begin{pmatrix} -6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -3 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 6 & 3 & 0 & -2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 3 & 6 & 0 & -1 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2 \end{pmatrix}, \quad \det = -2^7 3^6. \quad (\text{A.1})$$

**A.3. The flattened rank-7 minor.** Take columns  $a_0, \dots, a_6$  and rows

$$(a_0, u_0^2), (a_0, u_0 u_1), (a_0, u_0 u_2), (a_0, u_1^2), (a_0, u_1 u_2), (a_0, u_2^2), (a_1, u_0 u_1).$$

A row  $(\eta, u_k u_l)$  and a column  $\xi$  evaluate  $\bar{\Theta}_0^{(2)}(\xi \eta)$  on  $u_k u_l$ . The minor is

$$\begin{pmatrix} -6 & 0 & 0 & 6 & 0 & 0 & 0 \\ 0 & -3 & 0 & 3 & 0 & 3 & 0 \\ 0 & 0 & -3 & 3 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 & -3 & 3 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 6 \\ -3 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \det = -2^3 3^7. \quad (\text{A.2})$$

An exact rational-arithmetic verifier, `verify_exact_minors.py`, accompanies the source. The displayed nonzero determinants themselves are the proof certificates; the script only reproduces them.

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